PAPER_379 — MUGE Dual-Model 7-System Numeric Comparison: Compressed vs Resonance

Source: grok_share_11254865.txt, lines ~2700–3100
Parent document: "100. MUGE Compression cycle 3_11May2025.docx" (Grok full integration)
Session: 103 (Re-analysis pass — lines 2700–3100 read for first time)
CP4 Class: `DualModelMUGEComparisonCalculator` (CP4 #29)

Abstract

This paper presents a UQFF analysis of MUGE Dual-Model 7-System Numeric Comparison: Compressed vs Resonance, deriving compressed field equations and observational predictions within the Star-Magic/UQFF framework.

1. Overview

When Grok integrated both MUGE documents ("100." Compressed + "200." Resonance), it produced a full **side-by-side numeric comparison** of both models for all 7 canonical astrophysical systems. This paper captures that comparison table, the dominant-term analysis for each system in each model, and the key theoretical insight about when the models converge.

This data was NOT captured in PAPER_371 (resonance only) or PAPER_372 (compressed only).

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2. Full 7-System Numeric Comparison Table

System	Compressed MUGE g (m/s²)	Resonance MUGE g (m/s²)	Dominant Term (Compressed)	Dominant Term (Resonance)
Magnetar SGR 1745-2900	1.782×10^{39}	1.773×10^{-9}	Perturbation $(M \cdot \delta\rho/\rho)$	Fluid a_{fluid_freq}
Sagittarius A*	3.552×10^{20}	4.105×10^{29}	Fluid $\rho_{fl} \cdot V \cdot g$	Fluid a_{fluid_freq}
Tapestry of Blazing Starbirth	1.001×10^{27}	1.001×10^{27}	Fluid/Perturbation (tie)	Fluid a_{fluid_freq}

System	Compressed MUGE g (m/s²)	Resonance MUGE g (m/s²)	Dominant Term (Compressed)	Dominant Term (Resonance)
Westerlund 2	1.001×10^{27}	1.001×10^{27}	Fluid/Perturbation (tie)	Fluid a_{fluid_freq}
Pillars of Creation	2.001×10^{26}	2.001×10^{26}	Fluid	Fluid a_{fluid_freq}
Rings of Relativity	5.005×10^{25}	5.005×10^{25}	Fluid	Fluid a_{fluid_freq}
Student's Guide to the Universe	3.958×10^{14}	3.958×10^{14}	Fluid	Fluid a_{fluid_freq}

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3. Key Observations

3.1 SGR 1745-2900 — Extreme Divergence (48 Orders)

The magnetar case shows the **largest discrepancy** between the two models:

```
Compressed MUGE: g ≈ 1.782e39 m/s² (dominated by perturbation term)
  Perturbation = (M + M_DM) · (δρ/ρ + 3GM/r³)
               = (2.984e30 + 0) · (10⁻⁵ + 5.973e8)
               ≈ 1.782e39 m/s²

Resonance MUGE: g ≈ 1.773e-9 m/s² (dominated by fluid term)
  a_fluid_freq = f_fluid · E_vac,neb · V_sys / E_vac,ISM / c
               = 1.269e-14 × 7.09e-36 × 4.189e12 / 7.09e-37 / 3×10⁸
               ≈ 1.773e-9 m/s²
```

Physical interpretation: The compressed model's perturbation term is **unphysically large** for a magnetar (dense neutron star) — the $\delta\rho/\rho$ density perturbation applied to the full mass at neutron-star densities gives an unphysical result. The resonance model, which relies on vacuum energy density ratios and system volume, gives a physically plausible acceleration for a magnetar at scale.

This is evidence that the **Resonance MUGE is the more physically appropriate model for compact objects** (neutron stars, magnetars), while the Compressed MUGE is better suited to galactic/cosmological scales.

3.2 Sagittarius A* — Sgr A* Reversal (Resonance > Compressed by 9 Orders)

```
Compressed MUGE: g ≈ 3.552e20 m/s² (dominated by fluid: ρ_fl·V·g_local)
Resonance MUGE: g ≈ 4.105e29 m/s² (dominated by fluid: a_fluid_freq)
```

Both models are fluid-dominated for Sgr A*, but the resonance model gives a value 9 orders of magnitude **higher** — reflecting that the resonance model picks up the volume-scaled vacuum energy coupling ($E_{vac,neb} \cdot V_{sys}$) which is enormous for the SMBH volume $V_{sys} = 3.552 \times 10^{45}$ m³.

3.3 Star-Forming Regions and Beyond — Model Convergence

For the 5 remaining systems (Tapestry, Westerlund, Pillars, Rings, Student's Guide), **both models converge to the same value**. This is the empirical confirmation of the Cohesive UQFF principle (PAPER_378): in the star-forming/diffuse/large-volume regime, the resonance and compressed models give the same result — the fluid dynamics term dominates in BOTH models and is governed by the same physical inputs.

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4. System Parameters (Full Reference)

System 1: Magnetar SGR 1745-2900

```
M = 2.984e30 kg, r = 10⁴ m, t = 3.799e10 s, z = 0.0009
B = 10¹⁰ T, Bcrit = 10¹¹ T (B/Bcrit = 0.1 → 0.9 factor)
ρ_fluid = 10⁻¹⁵ kg/m³, V = 4.189e12 m³, g_local = 10 m/s²
M_DM = 0, δρ/ρ = 10⁻⁵
Resonance: I=10²¹ A, A=3.142e8 m², ω₁=10⁻³ rad/s, ω₂=-10⁻³ rad/s
vexp = 10³ m/s, ffluid = 1.269e-14 Hz
```

System 2: Sagittarius A*

```
M = 8.155e36 kg, r = 10¹² m, t = 3.786e14 s, z = 0.0009
B = 10⁻⁵ T, Bcrit = 10⁻⁴ T (B/Bcrit = 0.1 → 0.9 factor)
ρ_fluid = 10⁻²⁰ kg/m³, V = 3.552e45 m³, g_local = 10⁻⁵ m/s²
M_DM = 10³⁷ kg, δρ/ρ = 10⁻³
Resonance: I=10²³ A, A=2.813e30 m², ω₁=10⁻⁵, ω₂=-10⁻⁵ rad/s
vexp = 5×10⁶ m/s, ffluid = 3.465e-8 Hz
```

Systems 3–7 (Condensed)

```
Tapestry: M=10⁵ M⊙, r=10 pc, V=10⁵³ m³, ffluid=10⁻¹² Hz
Westerlund 2: Same as Tapestry (young cluster, stellar winds)
Pillars of Creation: M=10² M⊙, r=1 ly, V=10⁴⁸ m³, ffluid=10⁻¹⁰ Hz
Rings of Relativity: M=10⁶ M⊙, r=10 pc, V=10⁵⁴ m³, ffluid=10⁻⁹ Hz
Student's Guide: M≈10⁵³ kg (universe), r=10²⁶ m, V=10⁸⁰ m³, ffluid=10⁻¹⁸ Hz
```

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5. The Fluid Dynamics Universality Principle

A key result from this comparison: **in EVERY system in BOTH models, the fluid dynamics term ultimately dominates** (with the exception of the SGR1745 compressed case where the perturbation term wins).

Fluid dynamics universality:

```
For Resonance MUGE:   a_fluid_freq = f_fluid · E_vac,neb · V_sys / E_vac,ISM / c
For Compressed MUGE:   fluid_term   = ρ_fluid · V_sys · g_local
```

Both terms scale as $\propto V_{sys}$ — larger systems have proportionally larger fluid acceleration. This is consistent with Navier-Stokes turbulence being the governing dynamics at astrophysical scales (connecting to PAPER_369).

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6. Unit Test Reference Values

Derived from the Grok computation (validated against Grok's work-shown derivations):

System	Compressed g	Resonance g	Notes
SGR 1745-2900	1.782e39	**1.773e-9**	Resonance = physical; compressed unphysical
Sgr A*	3.552e20	**4.105e29**	Both fluid-dominated
Tapestry	1.001e27	1.001e27	Convergent
Westerlund	1.001e27	1.001e27	Convergent
Pillars	2.001e26	2.001e26	Convergent
Rings	5.005e25	5.005e25	Convergent
Student's Guide	3.958e14	3.958e14	Convergent

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7. Implementation

C++ (grok_share_11254865.txt, dual-model main()):

```cpp
for (const auto& sys : muge_systems) {
    double compressed_g = compute_compressed_MUGE(sys);
    double resonance_g  = compute_resonance_MUGE(sys, res_params);
    std::cout << "Compressed MUGE g for " << sys.name << ": " << compressed_g << " m/s2
    std::cout << "Resonance  MUGE g for " << sys.name << ": " << resonance_g  << " m/s2
}
```

Python CP4 class: `DualModelMUGEComparisonCalculator` (CP4 class #29)

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8. CP4 Class

Class: `DualModelMUGEComparisonCalculator`
Category: MUGE Validation / Comparison
Key method: `compute(dataset)` — takes 7-system catalog, returns compressed vs resonance table
References: PAPER_371, PAPER_372, PAPER_378

PAPER_379 | Session 103 | Star Magic UQFF Framework